

HS 312 – Introduction to Science and Technology Studies

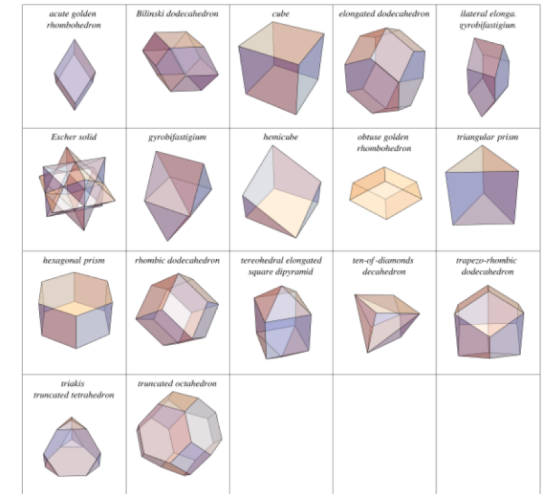
Lecture 13
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Reasonable disagreements

- Almost all participants in disputes have good reasons for their positions;
- **Symmetrical approach** attempts to show that labelling losing participants as unreasonable is unwarranted
- In the 1970s geophysicist Thomas Gold proposed an ‘abiogenic’ theory of the formation of hydrocarbons i.e. oil is not a fossil fuel but physical product
- Was it an irrational position ? Duhem-Quine thesis
- Even in maths like scientists and engineers, a theoretical position that is contrary to apparent evidence is maintained by some

Reasonable disagreements

- Disagreements are regularly and routinely managed and contained
- Most minority views are eventually excluded from public debates
- Extremely deviant views are marginalized
- For example; Leonard Euler's conjecture for polyhedral $\rightarrow V - E + F = 2$
- Proof was elegant but counter-examples were shown; so some wanted to derive universal relation that is true for all polyhedra; others mathematicians argued about looseness of the category polyhedra
- The meaning of polyhedra had to change in response to the proof and counter-examples



Experimenters' regress

- Experiments are normally thought to provide decisive evidence
- Experimental systems should be tools for producing differential responses based on different inputs
- At genuinely novel research fronts, experimenters do not know what their results will be
- Created to answer open questions that have not yet been answered
- **Experimenters' regress** – The experimental system is working when it gives the right answer, but one knows the right answer only after becoming confident in the experimental system

Experimenters' regress

- For example; measurement of gravitational waves in 1970s – Joseph Weber at Maryland had developed a large antenna but found many times larger than expected
- Others built equipment to replicate but found none; Weber had spent years calibrating his antenna
- Were their detectors working well? Or was Weber's device was measuring something else (an artefact) or simply erratic?
- Replication requires identical equipment – but that will reproduce same faults

Group Presentation

6	200110024	Banothu Aravind	Chapter 6	16	23B1212	Rahul Kumar	
	22B1000	Barre Varthika Reddy	The sex life of the whiptail	19-Feb	22B0903	Ratlan Sri Charan Singh Naik	
	22B1041	Bathala Shashank			17B030013	Rohit Auti	
	22B1042	Bhogi Siri			22B1028	Routhu Pavan Sekhar Naidu	

Sources

- Sismondo Sergio (2010). Questioning Functionalism in the Sociology of Science. Chapter 3, in *An Introduction to Science and Technology Studies*; second edition: Blackwell Publishing, Oxford UK.
- Harry Collins and Trevor Pinch (1998). The sex life of the whiptail lizard. Chapter 6, in *The Golem: what you should know about science*; second edition: Cambridge University Press, Cambridge UK.

Thank you